

Detecting Cyberbullying with NLP: A Comparative Study of Machine Learning Models

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Abstract—A text analysis pipeline for identifying and classifying cyberbullying in tweets is presented in this study. Standard text preprocessing, lexicon-based sentiment analysis (using TextBlob and VADER), and linguistic feature extraction (POS tagging and named entity recognition) are all integrated into the methodology. Three classification models (Random Forest, SVM, and Naive Bayes) are assessed, and a bag-of-words model is also built using TF-IDF. The findings show that the identification accuracy of content connected to cyberbullying is increased by combining statistical, sentiment, and grammatical elements.

I. INTRODUCTION

Cyberbullying has grown to be a concerning worldwide problem that impacts people of all ages and backgrounds. According to recent surveys, more than 37% of young people have already experienced cyberbullying in some capacity, and the number is still rising.

In order to detect and eliminate dangerous content, social media platforms usually rely on human moderators; however, this method is frequently costly, time-consuming, and prone to bias. Machine learning (ML) and natural language processing (NLP) offer viable, scalable solutions in response to the increasing number and complexity of cyberbullying incidents. Millions of texts can be quickly scanned in real time by AI-powered models, which can identify and classify harassing or malicious material before it does harm. Automating this procedure provides a proactive defence against cyberbullying by making extensive social media monitoring possible without constant human supervision.

II. DATASET AND PREPROCESSING

A. Dataset Description

The dataset used in this study contains a total of 47,692 tweets, with no missing values. Each tweet averages 136 characters in length, though some can be as long as 5,018 characters. The dataset includes two main columns:

- **tweet_text**: The text content of the tweet.
- **cyberbullying_type**: The classification label for each tweet.

The possible class labels for `cyberbullying_type` include `age`, `ethnicity`, `gender`, `religion`, `other_cyberbullying`, and `not_cyberbullying`. This variety of labels allows the dataset to capture multiple forms of harmful online behavior, providing a robust basis for multi-class classification tasks.

B. Text Preprocessing

Text Cleaning and Tokenisation

- **clean_tokens**: This column holds a list of tokens after the tweet has been processed. Processing steps include converting to lowercase, removing punctuation, filtering out stopwords, and then applying stemming and lemmatization.
- **clean_text**: This column is the reconstructed text where the tokens are joined back into a single string.

Parts-of-Speech (POS) Tagging and Chunking

- **pos_tags**: Here, each token is paired with a part-of-speech tag (e.g., NN for noun, JJ for adjective).
- **chunks**: Using a simple noun phrase grammar (NP: an optional determiner, any adjectives, and a noun), the tokens are grouped into chunks.

Sentiment Analysis

- **vader_sentiment**: This column shows sentiment scores calculated using the VADER sentiment analyzer. The dictionary includes neg, neu, pos, and compound.
- **textblob_sentiment**: This shows sentiment as computed by TextBlob, returning a tuple with polarity and subjectivity.

Named Entity Recognition (NER)

- **named_entities**: This column contains the entities extracted from the original tweet text using spaCy. Each entity is represented as a tuple (entity text, entity label).

The bag-of-words and TF-IDF matrices have shapes (47692, 53201), meaning there are 47,692 tweets processed and a vocabulary of 53,201 unique words after cleaning.

C. Exploratory Data Analysis

Word Cloud of Tweets

- **Chart**: A word cloud of the most common words (in larger font) across all cleaned tweets.
- **Observation**:
 - Words like “bully,” “school,” “idiot,” “muslim,” “nigger,” “fuck,” “rape,” and “gay” stand out, indicating strong or offensive language is common in this dataset.
 - This aligns with the nature of a cyberbullying corpus—many words relate to hate or harassment.

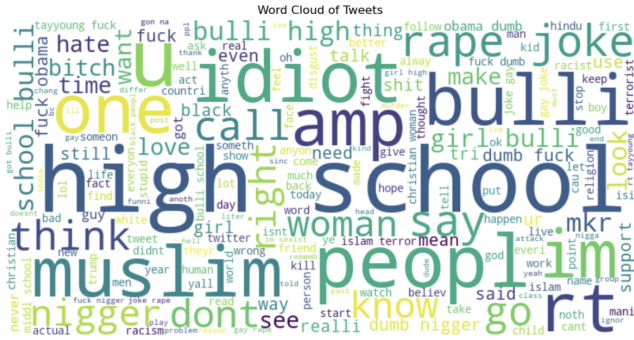


Fig. 1. Word Cloud of Frequent Terms

Top 20 Most Common Words

- **Chart:** A bar chart of the 20 most frequent words in the cleaned text.
- **Observation:**
 - “bulli” is at the top, likely because many tweets explicitly reference “bully/bullying.”
 - “school,” “fuck,” “like,” “nigga,” “peopl,” “amp,” “girl,” “u,” “hiphop,” “muslim,” “think,” “rape,” “man,” “gay,” “call,” “get,” and “one” are also frequent.
 - These words confirm the presence of profanity, slurs, and references to social contexts (e.g., “school,” “hiphop,” “muslim”).

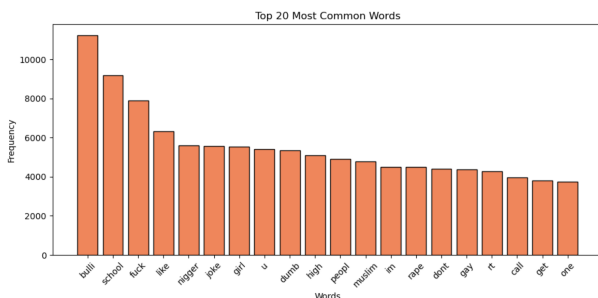


Fig. 2. Top 20 Most Common Words

D. Feature Extraction

The feature extraction process produces a comprehensive representation of each tweet by combining textual, sentiment, and linguistic attributes. Specifically, a TF-IDF matrix of shape (47692,5000) encodes the most discriminative 5,000 words across the dataset, as evidenced by sample terms such as "aa", "aalwuhaib", "abandon", and "ableist". These features capture term frequency while downweighting commonly used words, thereby emphasizing more informative vocabulary.

In parallel, a set of numeric features enriches the TF-IDF vectors with sentiment and linguistic information. VADER scores (`vader_neg`, `vader_neu`,

vader_pos, vader_compound) and TextBlob metrics (textblob_polarity, textblob_subjectivity) gauge the emotional tone of tweets. Meanwhile, token_count and entity_count quantify tweet length and named entities, respectively. POS-tag-based ratios (noun_ratio, verb_ratio, adj_ratio, adv_ratio) reflect the syntactic structure of each tweet. Descriptive statistics reveal that on average, tweets have a slight negative polarity (mean textblob_polarity -0.03), moderate subjectivity (0.39), and around 14 tokens per tweet. By integrating these diverse features, the dataset provides a rich foundation for downstream classification tasks.

III. CLASSIFICATION AND EVALUATION

Three models were implemented:

- **Naive Bayes:** Makes the assumption that each feature is independent and classifies using probability distributions.
- **Support Vector Machine (SVM):** Finds the best separating hyperplane in space with high dimensions.
- **Random Forest:** An ensemble approach that combines the forecasts of several different decision trees.

An 80/20 train-test split was used. Evaluation metrics include accuracy, macro F1-score, and weighted F1-score.

Table I presents the performance metrics of the three classifiers evaluated on the cyberbullying dataset. The SVM classifier achieves the highest accuracy at 82.33%, along with macro and weighted F1-scores of 82.45% and 82.30%, respectively. This suggests that SVM is particularly effective at handling the high-dimensional and sparse feature space generated by the TF-IDF and numeric features. In comparison, the Naive Bayes and Random Forest classifiers, while competitive, lag slightly behind. These results indicate that the SVM model is the most robust for the task of detecting and classifying cyberbullying content in social media data.

TABLE I
MODEL PERFORMANCE COMPARISON

Model	Accuracy	Macro F1-Score	Weighted F1-Score
Naive Bayes	0.765908	0.765155	0.763914
SVM	0.823252	0.824527	0.823013
Random Forest	0.801342	0.805743	0.804063

Figure 5 illustrates a performance comparison of the models.

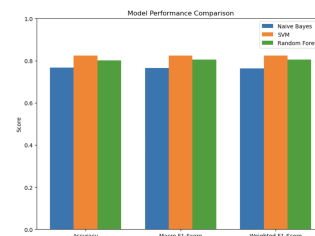


Fig. 3. Model performance comparison: Accuracy, Macro F1, and Weighted F1 scores.

IV. DISCUSSION AND CONCLUSION

Our research shows that the classification of cyberbullying content in tweets is greatly improved by combining textual, sentiment, and linguistic elements. According on the experimental data, the SVM classifier has the greatest accuracy (82.33%), weighted F1-score (82.30%), and macro F1-score (82.45%), consistently outperforming both Random Forest and Naive Bayes. This implies that the high-dimensional and sparse feature space produced by the combination of TF-IDF, sentiment, and grammatical characteristics is something that SVM excels at managing.

Given the complex and diverse nature of online abusive language, the SVM model's improved performance can be ascribed to its capacity to define a decision boundary in a complicated feature space. The Naive Bayes model, on the other hand, has the advantages of efficiency and simplicity, but it may oversimplify the interrelationships in the data due to its assumption of feature independence. Similarly, Random Forests did not capture the linear separability in this scenario as well as SVM, despite its robustness and capacity to handle non-linear interactions.

Despite these encouraging findings, a few drawbacks merit more examination. In the first place, our method mainly uses conventional TF-IDF representations, which may not adequately capture contextual subtleties in contrast to contemporary deep learning-based embeddings. Second, despite the relative category balance of our dataset, labelling ambiguities and biases may have an impact on model performance. Finally, further research is needed to determine whether our approach can be scaled to real-time applications and bigger, more varied datasets.

Future research could investigate more sophisticated methods, such transformer-based models (like BERT), which could improve comprehension of textual context and semantic subtleties. Furthermore, adding multimodal data or network-based features could improve the detection of cyberbullying even further. All things considered, our results demonstrate how well machine learning techniques—in particular, SVM—can recognise and classify harmful online conduct, opening the door for more automated and scalable cyberbullying detection systems.

V. LIMITATIONS

Despite the promising results, several limitations should be acknowledged:

- **Feature Representation:** Our method uses manually-engineered sentiment and syntactic features in conjunction with conventional TF-IDF representations. Even if it works effectively, this approach might not fully capture the contextual nuances of language as more recent deep learning-based embeddings (like BERT).
- **Data Annotation and Bias:** Despite the dataset's relative balance, labelling biases and ambiguities that come with hand annotation could affect it. The generalisability and performance of the model may be impacted by such discrepancies.
- **Scalability and Real-Time Application:** A predetermined dataset of tweets has been used to validate the existing system. More work is required to guarantee scalability and efficiency in dynamic contexts, and its performance in a real-time, large-scale monitoring scenario has not been proven.
- **Limited Contextual Analysis:** Contextual information from user interactions or conversation threads, which could offer additional insights regarding cyberbullying behaviour, is not included in the feature extraction process, which mostly concentrates on individual tweet content.

APPENDIX

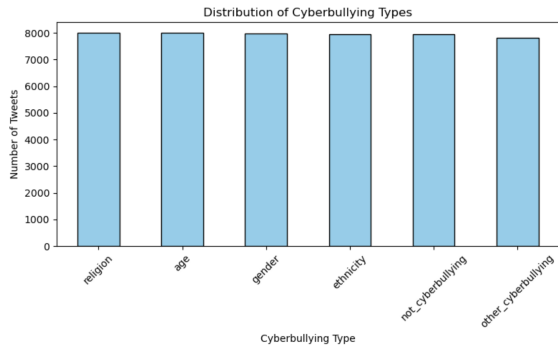


Fig. 4. Bar chart showing the number of tweets for each cyberbullying type

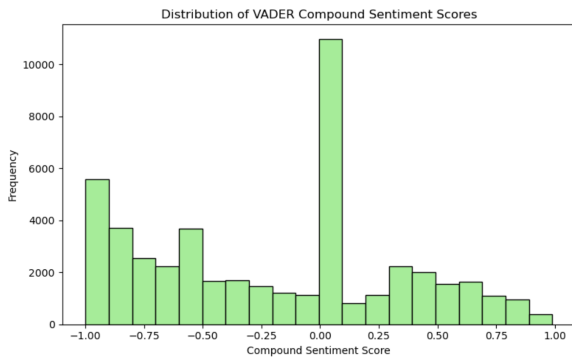


Fig. 5. A histogram of the VADER compound sentiment scores, which range from -1 (most negative) to +1 (most positive).